Practice workbook · Getting from A to B: subtracting position vectors

Finding the vector joining two points, or the position vector of a point, by subtracting or adding position vectors given in i, j or in letter vectors such as a and b.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-01

1. Question 1 · part a · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a only.

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1 Referred to a fixed origin O, the point A has position vector $(4\mathbf{i} + 3\mathbf{j})$ and the point B has position vector $(\mathbf{i} + 7\mathbf{j})$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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(Total for Question 1 is 4 marks)

1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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2. Question 7 · part a · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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2. Question 7 · part a · 2 marks · Working and answer

Working and answer

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3. Question 1 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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P 7 1 6 4 2 A 0 4 3 6

3. Question 1 · part a · 2 marks · Working and answer

Working and answer

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4. Question 10 · part a.i · 0 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

10

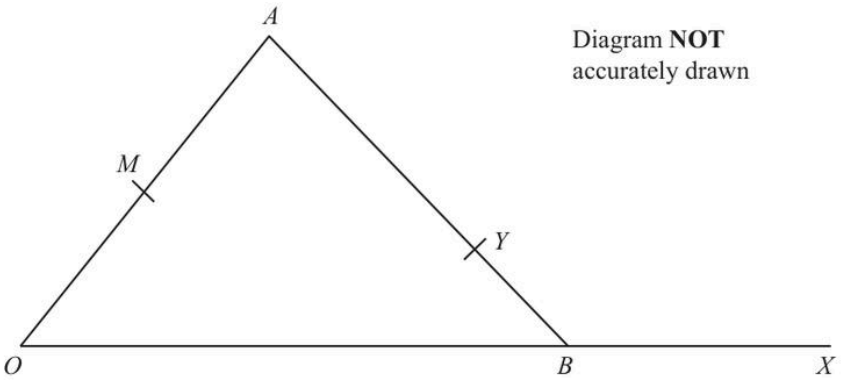


Diagram **NOT** accurately drawn

Figure 5

In Figure 5, $\overrightarrow{OA} = 2\mathbf{a}$, $\overrightarrow{OB} = 4\mathbf{b}$ and M is the midpoint of OA.

The point Y lies on AB such that $AY : YB = 3 : 1$

The point X lies on OB produced.

(a) Find as simplified expressions in terms of **a** and **b**

(i) $\overrightarrow{AB}$ (ii) $\overrightarrow{MY}$

(3)

The points M, Y and X are collinear.

(b) Find the ratio $OB : OX$

(5)

(c) Find the ratio of $(\text{Area } \Delta YBX) : (\text{Area } \Delta OAX)$

(3)

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4. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 0 marks

2024 NOV · Paper 02

Further practice · Attempt part a.i only.

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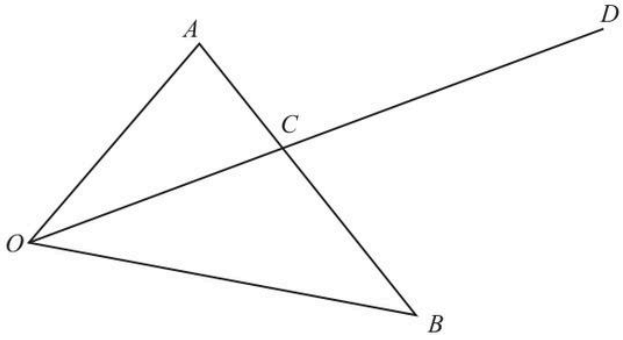


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Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$

(3)

(b) Find the area of triangle AOB

(4)

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b**

(5)

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P 7 3 5 8 7 A 0 2 0 3 2

5. Question 8 · part a.i · 0 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

(4)

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6. Question 4 · part b · 2 marks · Working and answer

Working and answer

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7. Question 9 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

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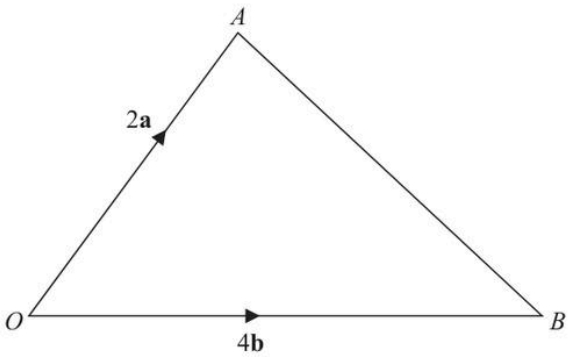


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows the triangle OAB with

$$\overrightarrow{OA} = 2\mathbf{a} \text{ and } \overrightarrow{OB} = 4\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(2)

The point P is the midpoint of AB

(b) Find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(2)

The point Q lies on OP such that $OQ : QP = 3 : 1$

(c) Find $\overrightarrow{AQ}$ as a simplified expression in terms of **a** and **b**

(3)

The point R lies on OB such that AQR is a straight line.

(d) Find in its simplest form the ratio $OR : RB$

(6)

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7. Question 9 · part a · 2 marks · Working and answer

Working and answer

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8. Question 8 · part a · 2 marks

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Further practice · Attempt part a only.

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Diagram **NOT** accurately drawn

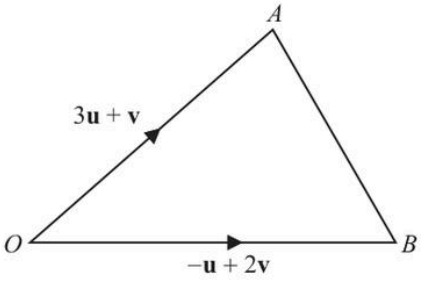


Figure 2

Figure 2 shows triangle OAB with

$$\overrightarrow{OA} = 3\mathbf{u} + \mathbf{v} \qquad \overrightarrow{OB} = -\mathbf{u} + 2\mathbf{v}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (2)

Point C is such that AC is parallel to OB

Given that $\overrightarrow{OC} = \mu\mathbf{v}$

(b) find the value of μ (4)

The lines OC and AB intersect at point X

Point D lies on $\overrightarrow{OC}$ such that

area of triangle BOX : area of triangle $BXD = 2 : 3$

Using a vector method,

(c) find $\overrightarrow{BD}$ as a simplified expression in terms of $\mathbf{u}$ and $\mathbf{v}$ (5)

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8. Question 8 · part a · 2 marks · Working and answer

Working and answer

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